

DIPOLE TYPE BY LENGTH
l = antenna rod length (m)

λ (wavelength) = $\frac{c}{f} = \frac{3 \times 10^8}{f}$ (m)		
<i>l</i> /λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D=1.5$
= 1/4	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
= 1/2	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73 \Omega$
= 1	Full-wave	ARB, $D=2.41$, $R_{\text{rad}}=199 \Omega$
other	Arbitrary	ARB formula

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors <i>small l</i> ≪ λ $\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$ Power density formulas <i>l</i>/λ < 1/50; far field	
Power density	$S(R, \theta) = S_{\text{max}} \sin^2 \theta$ (W/m ²)
Max pwr density	$S_{\text{max}} = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}$ (W/m ²)
<i>l</i> : rod length (m)	I_0 : peak current (A)
<i>R</i> : obs. distance (m)	θ : angle from dipole axis (rad)
<i>k</i> =2π/λ (rad/m)	$\eta_0=120\pi \approx 377 \Omega$
Total radiated power	$P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l / \lambda)^2$ (W)

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at <i>R</i> any <i>l</i>	
$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$	
<i>At</i> θ = 0, π: <i>bracket</i> is 0/0. <i>L'Hôpital</i> ⇒ <i>S</i> = 0 (no axial radiation). <i>TI-36X Pro alt</i> : evaluate <i>bracket</i> at θ = 0.001 <i>rad</i> → <i>tiny</i> → 0.	
Normalized pattern dimensionless $F(\theta) = S(\theta) / S_{\text{max}}$	

COMMON ANGLES (deg ↔ rad, trig values)

Conversion *multiply by $\pi/180$ or $180/\pi$*

$$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180} \qquad \theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$$

$^\circ$	rad	$\sin \theta$	$\cos \theta$	$\sin^2 \theta$	$\cos^2 \theta$
0	0	0	1	0	1
30	$\pi/6$	1/2	$\sqrt{3}/2$	1/4	3/4
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1/2	1/2
60	$\pi/3$	$\sqrt{3}/2$	1/2	3/4	1/4
90	$\pi/2$	1	0	1	0
180	π	0	-1	0	1

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

<i>N</i> elements, spacing <i>d</i> , equal phase ⇒ broadside (θ _{max} = π/2).	
Array factor (uniform, broadside) $k = 2\pi/\lambda$, $\gamma = kd \cos \theta$	
$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$; $F_a^{\text{max}} = N^2$ at θ = π/2	
Normalize $F_a^{\text{norm}} = F_a / N^2$	
HPBW (broadside, uniform) use <i>Nd</i>, not (<i>N</i>−1)<i>d</i> $\beta \approx 0.88 \frac{\lambda}{Nd}$ (rad) = 50.8° · $\frac{\lambda}{Nd}$	
Grating lobes appear if <i>d</i> > λ. Avoid.	

BEAM PARAMETERS — β, Ω_p, *D*

Normalize always start here $F(\theta) = S(\theta) / S_{\text{max}}$ (unitless) Beamwidth β at threshold <i>X</i> Set $F(\theta_X) = X$, solve for θ _X ; β = 2θ _X (symmetric)		
Threshold	$X = 10^{\text{dB}/10}$	θ _X (Gauss. short-cut)
HPBW (−3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$	any <i>X</i>	$F(\theta_X) = X$
Pattern solid angle Ω _p recognize pattern, look up $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$ (sr) <i>No</i> φ dep.: Ω _p = 2π ∫ ₀ ^π <i>F</i> (θ) sin θ <i>d</i> θ. <i>θ</i> is the integration variable — DON'T plug in a number for θ (e.g. θ _{1/2}). Integrate over θ ∈ [0, π].		
Pattern <i>F</i> (θ)	Ω _p (sr)	
Isotropic (<i>F</i> =1)	4π ≈ 12.57	
Hertzian sin ² θ	8π/3 ≈ 8.38	
Half-wave dipole	≈ 7.66	
Narrow Gauss <i>e</i> − <i>a</i> θ ²	π/ <i>a</i> = πθ _{1/2} ² /ln 2	
2-axis HPBW (aperture)	β _{xz} · β _{yz}	
Other (use integral)	numerical	

TI-36X Pro: [2nd] [d/dx] for $\int_{\square}^{\square}$, RAD mode, multiply by 2π.

Directivity <i>D</i> always <i>D</i> = 4π/Ω_p; common shortcuts below	
$D = \frac{4\pi}{\Omega_p}$ (unitless)	$D_{\text{dB}} = 10 \log_{10} D$
WORD TRAP: “direction” = θ _{max} (rad); “directivity” = <i>D</i> (unitless).	
Dipoles → COMMON DIPOLE PARAMS table. Aperture: $D \approx 4\pi A_p / \lambda^2$. 2-axis HPBW: $D \approx 4\pi / (\beta_{xz} \beta_{yz})$. Circular: $D \approx 4\pi / \beta^2$.	
Gain (if asked) ξ = rad. eff., 0 < ξ ≤ 1 $G = \xi D$ $G_{\text{dB}} = 10 \log_{10} G$	
Lossless / ideal antenna ⇒ <i>G</i> = <i>D</i> .	

COMMON DIPOLE PARAMETERS

Lossless (ξ=1): <i>G</i> = <i>D</i> .				
Type	<i>l</i>	<i>D</i>	<i>D</i> _{dB}	<i>R</i> _{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	< λ/50	1.5	1.76	80π ² (<i>l</i> /λ) ²
Half-wave	λ/2	1.64	2.15	73.2
Monopole	λ/4 (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199
Half-wave: <i>P</i> _{rad} = 36.6 <i>I</i> ₀ ² <i>W</i> . Monopole (λ/4 above ground): same patt. as half-wave but <i>P</i> _{rad} , <i>R</i> _{rad} halved.				
ANTENNA AREAS (<i>A</i> _e , <i>A</i> _p)				
Effective area antenna's RX cross section; <i>D</i> from BEAM PARAMS or table above $A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)				
Physical cross section wire dipole, diameter <i>d</i>; <i>l</i> see table above $A_p = l d$ (m ² , any dipole)				
Inverse: <i>D</i> from <i>A</i> _e $D = \frac{4\pi A_e}{\lambda^2}$ (unitless)				
<i>Thin wire</i> : <i>A</i> _e ≫ <i>A</i> _p (e.g., Hertzian <i>A</i> _e = 3λ ² /8π ≈ 0.119λ ²).				

FRISI TRANSMISSION FORMULA

For each antenna's *G*: if given by NAME (“half-wave dipole” etc.) → COMMON DIPOLE PARAMS (*G* = *D*, lossless). Else (dB/linear value given) → use as given (convert dB → linear).

Pwr density at RX, then RX pwr *G*_t, *G*_r linear gains; λ = *c*/*f*

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{); } P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$$

Equivalent: *P*_r = *S*_r *A*_{e,r}.

Solve for *R* max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$$

Equivalent form using *A*_e recv. area form
 $P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}$, $G_r = \frac{4\pi A_{e,r}}{\lambda^2}$

Free-space path loss: *L*_{fs} = (4π*R*/λ)², so *P*_r = *G*_t *G*_r *P*_t / *L*_{fs}.

Off-axis (patterns not aligned): multiply by *F*_t(θ_t, φ_t) *F*_r(θ_r, φ_r), both = 1 when aligned.

Recipe 1. λ = *c*/*f*. 2. Convert any dB gain → linear: *G* = 10^{*G*dB/10}.

3. Get *G*_t, *G*_r by antenna type: dipole→COMMON DIPOLE PARAMS; aperture→APERTURE box; arbitrary *F*→BEAM PARAMS. 4. Plug into Friis.

Use linear *G*, not dB. Convert FIRST.

dB ↔ LINEAR

Works for any power-like quantity (*G*, *D*, *P*_r/*P*_t, SNR, ...):

$$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}} \quad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: 3 dB ⇔ ×2, 10 dB ⇔ ×10. Add dB = multiply linear.

For *E*-field / amplitude use 20 log not 10 log (power ∝ *E*²).

NOISE & SNR

Noise power thermal noise into matched receiver $P_N = k_B T_{\text{sys}} B$ (W)
$k_B = 1.38 \times 10^{-23}$ J/K; <i>T</i> _{sys} system noise temp (K); <i>B</i> receiver bandwidth (Hz).
Signal-to-noise ratio $\text{SNR} = \frac{P_{\text{rec}}}{P_N}$ (unitless) $\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$
Typical comm. link needs SNR > 10 dB for usable signal.

APERTURE ANTENNAS (horn, dish)

*A*_p physical aperture area (m²); *A*_e effective area (m², = λ² *D* / 4π); β_{xz}, β_{yz} HPBW's in two planes (rad); *ℓ*, *d* aperture side / diameter (m).

Directivity (any shape, uniform illum.)
 $D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) ⇒ *A*_e ≈ *A*_p
Directivity from beamwidths any aperture
 $D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}}$ (use rad); circular: $D \approx 4\pi / \beta^2$

HPBW and <i>D</i> by aperture shape uniform illum.; β in rad			
Shape	<i>A</i> _p	HPBW (rad)	<i>D</i> (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x \ell_y$	$\beta_x \approx 0.88\lambda / \ell_x$ $\beta_y \approx 0.88\lambda / \ell_y$	$D \approx 4\pi / (\beta_x \beta_y)$ $= 4\pi \ell_x \ell_y / \lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda / \ell$	$D \approx 4\pi / \beta^2 = 4\pi \ell^2 / \lambda^2$
Circular (diam. <i>d</i>)	$\pi d^2 / 4$	$\beta \approx 1.02\lambda / d$	$D \approx 4\pi / \beta^2 = \pi^2 d^2 / \lambda^2$

Scaling (any ratio) $D \propto A_p / \lambda^2$; $\beta \propto \lambda / \sqrt{A_p}$
 $D_{\text{new}} = D_{\text{old}} \cdot \frac{A_{p,\text{new}}}{A_{p,\text{old}}} \cdot \left(\frac{f_{\text{new}}}{f_{\text{old}}} \right)^2$
 $\beta_{\text{new}} = \beta_{\text{old}} \cdot \sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}} \cdot \frac{f_{\text{old}}}{f_{\text{new}}}$

If a parameter is unchanged, its ratio = 1. Examples: double area → *D* ×2, β ×1/√2; double freq → *D* ×4, β ×1/2.

DIPOLE

<i>l</i> — antenna length (m) λ = <i>c</i> / <i>f</i> (m); <i>c</i> = 3 × 10 ⁸ m/s <i>I</i> ₀ — peak current (A) <i>k</i> = 2π/λ (rad/m) <i>R</i> — obs. dist. (m); θ from axis η ₀ = 120π ≈ 377 Ω <i>S</i> (<i>R</i> , θ) — pwr density (W/m ²)
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BEAM

<i>F</i> (θ) = <i>S</i> / <i>S</i> _{max} (unitless) <i>a</i> — coef. on θ ² in Gaussian β — HPBW (rad or °) Ω _p — pattern solid angle (sr) <i>R</i> = 4π/Ω _p — directivity ξ rad. eff.; <i>G</i> = ξ <i>D</i> <i>D</i> _{dB} = 10 log ₁₀ <i>D</i> ; <i>D</i> = 10 ^{<i>D</i>dB/10}

FRISI / dB

<i>P</i> _t — power transmitted (W) <i>P</i> _r — power received (W) <i>G</i> _t — transmit gain (linear) <i>G</i> _r — receive gain (linear) <i>S</i> _r = <i>G</i> _t <i>P</i> _t / (4π <i>R</i> ²) (W/m ²) <i>P</i> _r = <i>G</i> _t <i>G</i> _r (λ / 4π <i>R</i>) ² <i>P</i> _t <i>G</i> = 10 ^{<i>G</i>dB/10} 3 dB = ×2, 10 dB = ×10 <i>P</i> _N = <i>k</i> _B <i>T</i> _{sys} <i>B</i> (W)
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APERTURE

<i>A</i> _p — physical area (m ²) <i>A</i> _e = λ ² <i>D</i> / (4π) (m ²) <i>A</i> _e ≈ <i>A</i> _p for aperture $D \approx 4\pi A_p / \lambda^2$ (unitless) β ≈ 0.88λ / <i>ℓ</i> (rect/sq.) β ≈ 1.02λ / <i>d</i> (circular) 2 <i>A</i> _p ⇒ 2 <i>D</i> , β / √2

ARRAY

<i>N</i> — number of elements <i>d</i> — spacing (m) γ = <i>k</i> <i>d</i> cos θ <i>F</i> _a = sin ² (<i>N</i> γ/2) / sin ² (γ/2) <i>F</i> _a ^{max} = <i>N</i> ² at θ = π/2 <i>F</i> _a ^{norm} = <i>F</i> _a / <i>N</i> ² β ≈ 0.88λ / (<i>N</i> <i>d</i>) (broadside)
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